

Infinite Projections in Full Group C^* -Algebras of S -Arithmetic Groups

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Abstract

Let K be a number field, let Σ be a nonempty finite set of nonzero prime ideals of $\mathcal{O}_K$, and let $n \geq 4$. We prove that

$$C_{\max}^*(\mathrm{SL}_n(\mathcal{O}_{K,\Sigma}))$$

contains an infinite projection and a proper isometry. Hence it is not finite and is neither stably finite nor MF, although $\mathrm{SL}_n(\mathcal{O}_{K,\Sigma})$ is a finitely presented, residually finite property- (T) group. In particular this applies to $\mathrm{SL}_n(\mathbb{Z}[S^{-1}])$ for every nonempty finite set S of rational primes. More generally, if a countable discrete group G contains a property- (T) subgroup Γ and some $t \in G$ satisfies $t\Gamma t^{-1} \subsetneq \Gamma$, then the Kazhdan projection p_Γ is infinite in $C_{\max}^*(G)$; explicitly, $u_{t^{-1}}p_\Gamma + (1 - p_\Gamma)$ is a proper isometry. To the best of our knowledge, these are the first examples of countable discrete groups with a non-finite—and hence non-MF—full group C^* -algebra. The preparation of this manuscript was AI-assisted.

1 Introduction

All groups in this paper are countable and discrete. The faithful canonical trace makes the reduced group algebra $C_r^*(G)$ stably finite for every group G . By contrast, it was not known whether every full group C^* -algebra of a discrete group must be finite.

A projection in a C^* -algebra is *infinite* if it is Murray–von Neumann equivalent to a proper subprojection of itself. A unital C^* -algebra is finite precisely when every isometry is unitary, and it is stably finite if every matrix algebra over it is finite. Choi asked whether $C^*(G)$ can fail to be quasi-directly finite for a discrete group G [8, Question 1]. In the unital setting this asks whether $C^*(G)$ can contain a proper isometry. Milhøj and Rørdam later recorded the equivalent question whether a full group C^* -algebra can contain an infinite projection [13, Section 6]. The qualification “discrete” is essential: nondiscrete locally compact examples failing quasi-direct finiteness were already known [8, Section 3].

An MF C^* -algebra embeds into a norm ultraproduct of matrix algebras and is stably finite. In March 2026, Shulman noted that no group was known whose full or reduced group C^* -algebra is non-MF, and that stable finiteness of full group algebras was not known in general [15, Introduction].

Our main examples are the standard higher-rank S -arithmetic groups.

Theorem 1.1 (*S -arithmetic groups*). *Let K be a number field, let Σ be a nonempty finite set of nonzero prime ideals of $\mathcal{O}_K$, and let $\mathcal{O}_{K,\Sigma}$ be the ring of elements of K integral outside Σ . For every $n \geq 4$, the group*

$$G = \mathrm{SL}_n(\mathcal{O}_{K,\Sigma}),$$

viewed as a discrete group, is finitely presented, linear, residually finite, and has property (T), while $C_{\max}^*(G)$ contains a proper isometry and an infinite projection. Consequently, $C_{\max}^*(G)$ is not finite and is neither stably finite nor MF.

For $K = \mathbb{Q}$, write $\mathbb{Z}[S^{-1}] = \mathbb{Z}[p^{-1} : p \in S]$. The subgroup and compressor behind Theorem 1.1 are especially transparent: for $p \in S$, put

$$\Gamma = \left\{ \begin{pmatrix} A & x \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_{n-1}(\mathbb{Z}), x \in \mathbb{Z}^{n-1} \right\}, \quad t = \mathrm{diag}(p, \dots, p, p^{-(n-1)}).$$

Then $\Gamma \cong \mathbb{Z}^{n-1} \rtimes \mathrm{SL}_{n-1}(\mathbb{Z})$ has property (T) and

$$t\Gamma t^{-1} = p^n \mathbb{Z}^{n-1} \rtimes \mathrm{SL}_{n-1}(\mathbb{Z}) \subsetneq \Gamma.$$

The arithmetic theorem is an instance of the following general mechanism.

Theorem 1.2 (Kazhdan compression criterion). *Let G contain a property-(T) subgroup Γ . If some $t \in G$ satisfies*

$$t\Gamma t^{-1} \subsetneq \Gamma,$$

then the Kazhdan projection p_Γ is infinite in $C_{\max}^(G)$, and $C_{\max}^*(G)$ contains a proper isometry. Consequently, $C_{\max}^*(G)$ is not finite and hence is neither stably finite nor MF.*

The mechanism is elementary. In every representation, the Kazhdan projection $p = p_\Gamma$ projects onto the Γ -fixed vectors. The subgroup inclusion implies

$$pu_{t^{-1}}p = u_{t^{-1}}p.$$

Thus $v = u_{t^{-1}}p$ is an isometry in the corner $pC_{\max}^*(G)p$, with

$$v^*v = p, \quad vv^* = u_{t^{-1}}pu_t \leq p.$$

The quasi-regular representation on $\ell^2(G/\Gamma)$ detects that the last inequality is strict. Hence p is infinite, and $v + (1 - p)$ is a proper isometry with initial projection 1.

Every residually finite group is MF as a group [7, Proposition 2.13]. Theorem 1.1 therefore shows, even among finitely presented residually finite Kazhdan groups, that the converse to

$$C_{\max}^*(G) \text{ is MF} \implies G \text{ is MF}$$

fails.

Wassermann used Kazhdan projections in full group C^* -algebras for extension-theoretic obstructions [16]. Bekka obtained finite-dimensional and tracial rigidity results for full group algebras of arithmetic groups [1, 2]; these do not imply failure of finiteness. One-sided subgroup compression also produces isometries in Hecke C^* -algebras and semigroup crossed products [12]. Those isometries live in Hecke completions rather than in the full group C^* -algebra of the ambient discrete group. Here property (T) supplies the universal fixed-vector projection inside the maximal group C^* -algebra itself.

Section 2 proves the criterion and locates its defect projection. Section 3 proves the arithmetic theorem. Section 4 records the resulting separation of maximal, finite-dimensional, and reduced completions.

2 The compression argument

We first recall two standard facts in the form needed below.

Lemma 2.1 (Maximal subgroup inclusion). *If $H < G$, the canonical map $\mathbb{C}[H] \rightarrow \mathbb{C}[G]$ extends to an injective $*$ -homomorphism*

$$C_{\max}^*(H) \longrightarrow C_{\max}^*(G).$$

Proof. Restriction of representations gives $\|x\|_{C_{\max}^*(G)} \leq \|x\|_{C_{\max}^*(H)}$ for $x \in \mathbb{C}[H]$. Conversely, if π is a unitary representation of H , then the restriction to H of $\text{Ind}_H^G \pi$ contains π on the fibre over $H \in G/H$. Hence

$$\|\pi(x)\| \leq \|\text{Ind}_H^G \pi(x)\| \leq \|x\|_{C_{\max}^*(G)}.$$

Taking the supremum over π proves the claim. $\square$

Proposition 2.2 (The Kazhdan projection). *If Γ has property (T), there is a central projection $p_\Gamma \in C_{\max}^*(\Gamma)$ such that, in every unitary representation π of Γ , $\pi(p_\Gamma)$ is the orthogonal projection onto $\mathcal{H}^\Gamma$.*

More precisely, let $S \subset \Gamma$ be a nonempty finite symmetric Kazhdan set with Kazhdan constant $0 < \kappa \leq 2$, and put

$$M = \frac{1}{|S|} \sum_{s \in S} u_s, \quad L = \frac{1+M}{2}, \quad r = 1 - \frac{\kappa^2}{4|S|} < 1.$$

Then $L^j \rightarrow p_\Gamma$ in maximal norm and

$$\|L^j - p_\Gamma\| \leq r^j.$$

Proof. Since S is symmetric, $M = M^*$ and $\|M\| \leq 1$, so L is a positive contraction. If $\xi \perp \mathcal{H}^\Gamma$, then

$$\|\xi\|^2 - \langle \pi(M)\xi, \xi \rangle = \frac{1}{2|S|} \sum_{s \in S} \|\pi(s)\xi - \xi\|^2 \geq \frac{\kappa^2}{2|S|} \|\xi\|^2.$$

Thus $\pi(L)$ is the identity on $\mathcal{H}^\Gamma$ and has norm at most r on its orthogonal complement, uniformly over all representations. It follows that L^j converges in $C_{\max}^*(\Gamma)$ with the stated estimate, and its limit has the required universal property. The invariant-vector projection commutes with $\pi(\Gamma)$ in every representation, so the limit is central. See also [11]. $\square$

Proof of Theorem 1.2. By Lemma 2.1, regard $p = p_\Gamma$ as an element of $A = C_{\max}^*(G)$. Put

$$v = u_{t-1}p, \quad q = u_{t-1}pu_t.$$

Let U be any unitary representation of G , and let $P = U(p)$ be the projection onto $\mathcal{H}^\Gamma$. For $\xi \in \mathcal{H}^\Gamma$ and $\gamma \in \Gamma$,

$$U_\gamma U_{t-1} \xi = U_{t-1} U_{t\gamma t^{-1}} \xi = U_{t-1} \xi.$$

Hence $U_{t-1} \mathcal{H}^\Gamma \subseteq \mathcal{H}^\Gamma$, so

$$PU_{t-1}P = U_{t-1}P.$$

Since this holds in every representation,

$$pu_{t-1}p = u_{t-1}p = v.$$

Therefore $v \in pAp$ and

$$v^*v = p, \quad vv^* = q \leq p.$$

To prove strictness, consider the quasi-regular representation

$$\rho : G \curvearrowright \ell^2(G/\Gamma).$$

The ranges of $\rho(p)$ and $\rho(q)$ are the Γ -fixed and $(t^{-1}\Gamma t)$ -fixed subspaces, respectively. Since $\Gamma \subsetneq t^{-1}\Gamma t$, choose $h \in t^{-1}\Gamma t \setminus \Gamma$. The vector δ_Γ is Γ -fixed, whereas

$$\rho(h)\delta_\Gamma = \delta_{h\Gamma} \neq \delta_\Gamma.$$

Thus $\rho(q) \neq \rho(p)$, and consequently $q < p$ in A . This proves that p is infinite. Finally, with

$$d = p - q, \quad w = v + (1 - p),$$

the cross terms vanish and

$$w^*w = 1, \quad ww^* = 1 - d < 1.$$

Hence w is a proper isometry. An MF C^* -algebra is stably finite [4], so A is not MF. $\square$

The proof uses property (T) only to obtain a projection whose image in every representation of G is the projection onto the Γ -fixed vectors. Whenever such a projection exists, the same argument applies.

The identities above also determine exactly where the defect $d = p - q$ is visible.

Proposition 2.3 (The compression defect). *Under the hypotheses of Theorem 1.2, let p, v, q, d be as in its proof.*

1. *Every $*$ -homomorphism from $C_{\max}^*(G)$ to a finite unital C^* -algebra annihilates d . Hence every finite-dimensional representation and every tracial GNS representation annihilates the closed ideal generated by d . In particular, $C_{\max}^*(G)$ is not residually finite-dimensional and has no faithful tracial state.*
2. *The elements p, v, q, d lie in the kernel I of $C_{\max}^*(G) \rightarrow C_r^*(G)$. The projection p is infinite but not properly infinite, and the C^* -algebra generated by v in the corner with unit p is isomorphic to the Toeplitz algebra $\mathcal{T}$, with defect projection $d = p - vv^*$.*
3. *The quasi-regular representation $\lambda_{G/\Gamma}$ does not annihilate d . Consequently, the completion associated with $\lambda_G \oplus \lambda_{G/\Gamma}$ is strictly larger than the reduced completion, although it may coincide with the maximal completion.*

Proof. Let $\Phi : C_{\max}^*(G) \rightarrow B$, with B finite and unital. Passing to the finite corner $\Phi(1)B\Phi(1)$, assume that Φ is unital. The partial isometry $\Phi(v)$ implements

$$\Phi(p) \sim \Phi(q) \leq \Phi(p).$$

Finiteness forces $\Phi(p) = \Phi(q)$, so $\Phi(d) = 0$. This covers finite-dimensional representations. For a tracial state τ one also has $\tau(d) = \tau(v^*v) - \tau(vv^*) = 0$; since d is a projection, its image in the GNS representation is zero. As $d \neq 0$, the asserted failures of residual finite dimensionality and faithful traciality follow.

Strict compression forces Γ to be infinite. The restriction of the left regular representation of G to Γ has no invariant vector, and hence annihilates p . Thus $p, v, q, d \in I$. The relation $v^*v = p$ and the strict inequality $vv^* < p$ make p infinite. The augmentation character sends p to 1, so p is not properly infinite. Coburn's theorem [9] identifies the algebra generated by the nonunitary isometry v in $pC_{\max}^*(G)p$ with $\mathcal{T}$.

Finally, the proof of Theorem 1.2 shows that $\lambda_{G/\Gamma}(d) \neq 0$, proving the last assertion. $\square$

3 S-arithmetic groups

Proof of Theorem 1.1. Set $m = n - 1$. Choose $\mathfrak{p} \in \Sigma$. Since the ideal class group of K is finite, there are $h \geq 1$ and $\alpha \in \mathcal{O}_K$ such that

$$\mathfrak{p}^h = (\alpha).$$

Thus α is a nonunit of $\mathcal{O}_K$ and, since $(\alpha) = \mathfrak{p}^h$ is supported in Σ , a unit of $\mathcal{O}_{K,\Sigma}$.

Inside $G = \mathrm{SL}_n(\mathcal{O}_{K,\Sigma})$ consider

$$\Gamma = \left\{ \begin{pmatrix} A & x \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_m(\mathcal{O}_K), x \in \mathcal{O}_K^m \right\} \cong \mathcal{O}_K^m \rtimes \mathrm{SL}_m(\mathcal{O}_K).$$

By Borel–Harish-Chandra, applied after restriction of scalars from K to $\mathbb{Q}$, the diagonal archimedean embedding realizes Γ as a lattice in

$$\prod_{v|\infty} (K_v^m \rtimes \mathrm{SL}_m(K_v)).$$

See [5]. Each factor has property (T) for $m \geq 3$ [3, Corollary 1.4.16], and property (T) passes to lattices [3, Theorem 1.7.1]. Hence Γ has property (T).

Put

$$t = \mathrm{diag}(\alpha I_m, \alpha^{-m}) \in G.$$

For $A \in \mathrm{SL}_m(\mathcal{O}_K)$ and $x \in \mathcal{O}_K^m$, direct multiplication gives

$$t \begin{pmatrix} A & x \\ 0 & 1 \end{pmatrix} t^{-1} = \begin{pmatrix} A & \alpha^{m+1}x \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} A & \alpha^n x \\ 0 & 1 \end{pmatrix}.$$

Consequently,

$$t\Gamma t^{-1} = \alpha^n \mathcal{O}_K^m \rtimes \mathrm{SL}_m(\mathcal{O}_K) \subsetneq \Gamma,$$

where strictness follows because α is a nonunit. Theorem 1.2 gives the proper isometry and infinite projection.

For completeness, the remaining group-theoretic properties are standard. The standard S-arithmetic lattice theorem realizes G diagonally as a lattice in

$$\prod_{v|\infty} \mathrm{SL}_n(K_v) \times \prod_{\mathfrak{q} \in \Sigma} \mathrm{SL}_n(K_{\mathfrak{q}})$$

[6]. The local-field theorem for SL_n and lattice inheritance give property (T) [3, Theorems 1.4.15 and 1.7.1]. The finiteness theorem of Borel–Serre gives finite presentability [6], and linearity is immediate.

To see residual finiteness directly, let $g \neq 1$ and choose a nonzero entry $x \in \mathcal{O}_{K,\Sigma}$ of $g - I_n$. A nonzero element of the Dedekind domain $\mathcal{O}_{K,\Sigma}$ lies in only finitely many maximal ideals, so one can choose a nonzero prime ideal $\mathfrak{q} \notin \Sigma$ with $x \notin \mathfrak{q}\mathcal{O}_{K,\Sigma}$. Reduction modulo $\mathfrak{q}$ maps G to the finite group $\mathrm{SL}_n(\mathcal{O}_K/\mathfrak{q})$ and does not kill $g - I_n$. $\square$

In the rational case the class-group step disappears. If $p \in S$, then $\alpha = p$ and the displayed calculation reads

$$\mathrm{diag}(pI_{n-1}, p^{-(n-1)}) \begin{pmatrix} A & x \\ 0 & 1 \end{pmatrix} \mathrm{diag}(p^{-1}I_{n-1}, p^{n-1}) = \begin{pmatrix} A & p^n x \\ 0 & 1 \end{pmatrix}.$$

The affine property-(T) group and its finite-index self-compressions also appear in [10, Proposition 1.1].

4 Finite-dimensional approximation

For a group G and $x \in \mathbb{C}[G]$, set

$$\|x\|_{\text{fd}} = \sup\{\|\pi(x)\| : \pi \text{ is a finite-dimensional unitary representation of } G\}.$$

Finite-dimensional exotic group completions have been studied previously; see [17, 14]. The compression defect makes both quotient kernels below explicit.

Proposition 4.1. *Let G be residually finite and satisfy Theorem 1.2. Then*

$$\|x\|_r \leq \|x\|_{\text{fd}} \leq \|x\|_{\text{max}} \quad (x \in \mathbb{C}[G]),$$

and $\|\cdot\|_{\text{fd}}$ is a C^* -norm on $\mathbb{C}[G]$. Its completion $C_{\text{fd}}^*(G)$ is residually finite-dimensional and stably finite. Both canonical quotients

$$C_{\text{max}}^*(G) \longrightarrow C_{\text{fd}}^*(G) \longrightarrow C_r^*(G)$$

are proper: the first kernel contains

$$d = p_\Gamma - u_{t-1} p_\Gamma u_t,$$

and the second kernel contains the image of p_Γ .

Proof. Let $0 \neq x = \sum_{g \in F} a_g u_g \in \mathbb{C}[G]$ with F finite. Residual finiteness gives a finite quotient of G that is injective on F . The image of x in the group algebra of this quotient is nonzero, and the regular representation of the quotient is faithful. Thus $\|x\|_{\text{fd}} > 0$.

To compare with the reduced norm, let $\xi \in \ell^2(G)$ be finitely supported. There is a finite quotient $\theta : G \rightarrow Q$ that is injective on

$$F \cup \text{supp}(\xi) \cup F \text{supp}(\xi).$$

Transporting ξ to $\ell^2(Q)$ then preserves both its norm and the norm of its image under x . Taking the supremum over such ξ gives $\|x\|_r \leq \|x\|_{\text{fd}}$.

The product of all finite-dimensional representations realizes $C_{\text{fd}}^*(G)$ as a C^* -subalgebra of a product of matrix algebras, so it is residually finite-dimensional and stably finite. Proposition 2.3 gives $0 \neq d \in \ker(C_{\text{max}}^*(G) \rightarrow C_{\text{fd}}^*(G))$. The image of p_Γ in $C_{\text{fd}}^*(G)$ is nonzero because the augmentation character sends it to 1, whereas its image in $C_r^*(G)$ is zero. Hence the second quotient is proper. $\square$

Thus every group in Theorem 1.1 has a canonical residually finite-dimensional, stably finite exotic completion strictly between its maximal and reduced group algebras.

The construction of Proposition 2.2 also gives finite-support witnesses for the norm loss. Namely,

$$a_j = L^j - u_{t-1} L^j u_t \in \mathbb{C}[G]$$

satisfies

$$\|a_j - d\|_{\text{max}} \leq 2r^j, \quad \|a_j\|_{\text{fd}} \leq 2r^j, \quad \|\lambda_{G/\Gamma}(a_j)\| \geq 1 - 2r^j.$$

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